Bidirectional Token-Based Medium Access Control for Wireless Network-on-Chip Architectures

***Abstract*—Wireless Network-on-Chip (WiNoC) performance depends partly on how radio hubs share the wireless medium. With unidirectional token circulation, a request that arrives after the token has departed may remain pending for nearly an entire rotation. This study examines a bidirectional token-based MAC (BMAC) that permits a local return to the preceding hub when the backtracking condition is met. A simplified service-step analysis relates token access delay to ring size and illustrates the potential benefit of choosing the shorter traversal direction; this idealized case is distinguished from the local mechanism. Cycle-accurate Noxim experiments compare BMAC with a token-ring baseline in a 64-core mesh containing 16 radio hubs. Random, shuffle, and transpose workloads show higher saturation throughput, whereas hotspot traffic produces little throughput improvement. The largest reported throughput ratio is approximately $1.84\times$, observed for shuffle traffic. The results indicate that the benefit depends on the communication pattern rather than applying uniformly across workloads.**

***Index Terms*—Bidirectional Token, Medium Access Control, Wireless, Network-on-Chip, Protocol, Architecture.**

I. Introduction

Hybrid WiNoCs combine a wired fabric with wireless paths that can bypass multiple router hops. This arrangement is useful when communication spans distant regions of a chip. Recent topology and mapping studies examine how the placement and use of wireless links affect communication cost [1]–[3]. Advances in integrated antennas and transceivers also support continued investigation of wireless on-chip and in-package links [4]–[7]. A shared radio channel, however, requires arbitration among the hubs that wish to transmit.

Token-based arbitration assigns transmission permission to the hub that currently owns the token. A logical ring gives each hub a position in the service order and avoids simultaneous transmissions on the shared channel. Token schemes have therefore been studied in several WiNoC designs [8]–[10]. Their access delay depends on the service order: a hub with a new request may need to wait while the token visits other hubs. Alternative approaches include time-reversal MAC and pulse-based terahertz communication [5], [10], while traffic-aware arbitration provides a broader motivation for matching access decisions to workload behavior [11].

DiTomaso et al. used token arbitration in iWISE, which supplements router connections with wireless express paths to reduce communication hop count [12]. Their subsequent adaptive WiNoC work reallocates wireless resources according to bandwidth demand and reports improvements of 21% in energy efficiency and 37% in throughput [13]. These results concern that architecture and are separate from the BMAC evaluation presented here.

Hierarchical small-world WiNoCs also use token ownership to coordinate access to millimeter-wave express links [14], [15]. In these systems, a transmitting hub releases the token after completing its permitted flits, allowing another hub to use the medium. The wireless express path reduces reliance on repeated wired forwarding. More recent topology and mapping research continues to examine the interaction between wireless access and network organization [1], [3], [4].

Bidirectional MAC has also been described in earlier BMAC and distance-aware DA+BMAC publications [16], [17]. The present paper examines local token backtracking through a service-step model and a comparison with a unidirectional baseline. The evaluation below focuses on token access and throughput in a fixed mesh configuration, with particular attention to the distinction between ideal bidirectional traversal and local backtracking.

II. Bidirectional Token-Based MAC Protocol

To isolate the effect of traversal direction, consider N radio hubs sharing a single token. Token ownership permits a hub with a pending wireless request to transmit. We measure access delay using W, the cycles from a request becoming pending until its hub receives the token. The following assumptions define the simplified model:

- N radio hubs are organized in a logical ring.
- The token moves in discrete *service steps*. At each step, the token arrives at a hub, which either (i) transmits for a bounded holding time of T_h cycles if it has a request or (ii) immediately forwards the token (incurring forwarding time T_f cycles).
- For analytical tractability, we treat the per-hop token advance time as a constant

$$T_s \triangleq T_f + \rho T_h, \tag{1}$$

  where $\rho \in [0, 1]$ represents the average fraction of hubs that hold the token for data transmission. In a saturated or heavily loaded regime, $\rho \approx 1$ and thus $T_s \approx T_f + T_h$. In a lightly loaded regime, $\rho \ll 1$ and $T_s \approx T_f$.

Let $d \in \{0, 1, \ldots, N-1\}$ denote the *token distance* from the current token position to the hub where a request arrives, measured as the number of hops in the token-traversal direction for the baseline unidirectional ring.

1) Baseline Unidirectional Token Ring: In a conventional unidirectional token ring, the request must wait until the token advances d hops. Therefore, the waiting time is

$$W_{\text{base}}(d) = d\,T_s. \tag{2}$$

a) Worst-case waiting time.: The worst case occurs when the token has just passed the requesting hub, i.e., $d = N - 1$:

$$W_{\text{base}}^{\max} = (N-1)\,T_s. \tag{3}$$

b) Average waiting time (uniform request arrival).: If request arrivals are uniformly distributed over the ring position of the token, then d is uniformly distributed on $\{0, \ldots, N-1\}$ and

$$\mathbb{E}[W_{\text{base}}] = \mathbb{E}[d]\,T_s = \frac{N-1}{2}\,T_s. \tag{4}$$

Equations (3)–(4) show that token waiting time scales linearly with N, creating a scalability bottleneck for large WiNoCs.

BMAC introduces bidirectional token traversal so that the token can *backtrack* to serve a request from the immediately preceding neighborhood when a neighbor-aware trigger condition is satisfied.

To bound the potential benefit, we model an *idealized* bidirectional token in which, upon request arrival, the token can approach the requesting hub from the shorter direction on the ring. Under this model, the effective distance becomes

$$d_{\min} \triangleq \min(d,\, N-d). \tag{5}$$

Hence,

$$W_{\text{bmac}}^{\star}(d) = d_{\min}\,T_s. \tag{6}$$

c) Worst-case bound.: The worst-case effective distance is $\lfloor N/2 \rfloor$, which yields

$$(W_{\text{bmac}}^{\star})^{\max} = \left\lfloor \frac{N}{2} \right\rfloor T_s, \tag{7}$$

which represents an approximately $2\times$ reduction compared with $W_{\text{base}}^{\max}$ for large N.

d) Average waiting time (uniform request arrival).: For uniform d, the expected value of $d_{\min}$ is

$$\mathbb{E}[d_{\min}] = \begin{cases} \frac{N}{4}, & N \text{ even}, \\ \frac{N^2-1}{4N}, & N \text{ odd}, \end{cases} \tag{8}$$

and thus

$$\mathbb{E}[W_{\text{bmac}}^{\star}] = \mathbb{E}[d_{\min}]\,T_s \approx \frac{N}{4}\,T_s. \tag{9}$$

Combining (4) and (9) shows that the idealized bidirectional token reduces the average waiting time by approximately $2\times$:

$$\frac{\mathbb{E}[W_{\text{base}}]}{\mathbb{E}[W_{\text{bmac}}^{\star}]} \approx \frac{(N-1)/2}{N/4} \approx 2. \tag{10}$$

In the proposed Bidirectional Token, backtracking is not always invoked; it occurs only when a neighbor-aware condition is satisfied, such as when the next request corresponds to the previous adjacent hub. Let $\alpha \in [0, 1]$ denote the probability that the Bidirectional Token backtracking trigger is activated for a random request event.

For illustration, we interpolate between baseline service and ideal shorter-direction service. This heuristic assumes the same request-distance distribution in both cases; it is not a derived waiting-time guarantee for the local backtracking controller:

$$\mathbb{E}[W_{\text{bmac}}] = \alpha\,\mathbb{E}[W_{\text{bmac}}^{\star}] + (1-\alpha)\,\mathbb{E}[W_{\text{base}}]. \tag{11}$$

Substituting (4) and (9) yields

$$\mathbb{E}[W_{\text{bmac}}] \approx \alpha\left(\frac{N}{4}T_s\right) + (1-\alpha)\left(\frac{N-1}{2}T_s\right). \tag{12}$$

In this interpolation, increasing α moves the estimated delay toward the ideal shorter-direction value. A local trigger does not establish that every activated request receives shortest-path service, and repeated reversals can affect other pending requests. Consequently, neither the ideal bound nor the mixture establishes a worst-case bound for the implemented BMAC. The shuffle results are compatible with a locality-based explanation, but the experiments do not directly measure α or validate this interpolation.

TABLE I
SIMULATION SETUP FOR THE 8×8 (64-CORE) MM-WAVE MESH-WINOC ARCHITECTURE WITH 16 RADIO HUBS.

Parameter	Description
Network Size	8×8 (64 cores)
Number of Radio Hubs	16
Synthetic Traffic	Random, Shuffle, Transpose, Hotspot
Simulation Time	100,000 cycles
Technology	65 nm
Clock Frequency	1 GHz
Radio Access Control	Baseline Token Ring [21], Bidirectional Token
Flit Size	32 bits
Routing Algorithm	XY
Wireless Data Rate	16 Gbps (mm-wave)

III. PERFORMANCE EVALUATION

The experiments use Noxim to compare the two access protocols in the same mesh configuration. The wired layer contains 64 processing cores arranged as an 8×8 grid, with 16 radio hubs forming a 4×4 overlay. Table I lists the remaining parameters, and Figure 1 depicts the topology. Each run lasts 100,000 cycles; measurements exclude the initial 1,000 warm-up cycles. Cycle-accurate evaluation is also used in recent NoC studies [18]–[20]. Here, the comparison is restricted to the four synthetic workloads listed in the table.

Figure 2 plots average packet latency against packet injection rate (PIR) for random, shuffle, transpose, and hotspot traffic. Shuffle traffic shows the clearest separation between the protocols: the reported maximum supported load rises from 0.016 flit/cycle/tile with the baseline to 0.029 flit/cycle/tile with BMAC. Local backtracking can serve recently passed hubs without waiting for another full rotation, offering a possible explanation for the improvement under this communication pattern.

The hotspot workload concentrates demand at a central destination and therefore tests an uneven traffic distribution. The hotspot destination is configured to be 80% more likely

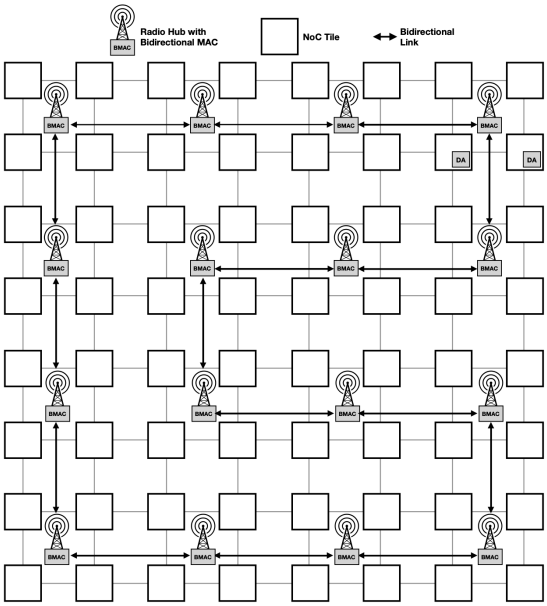
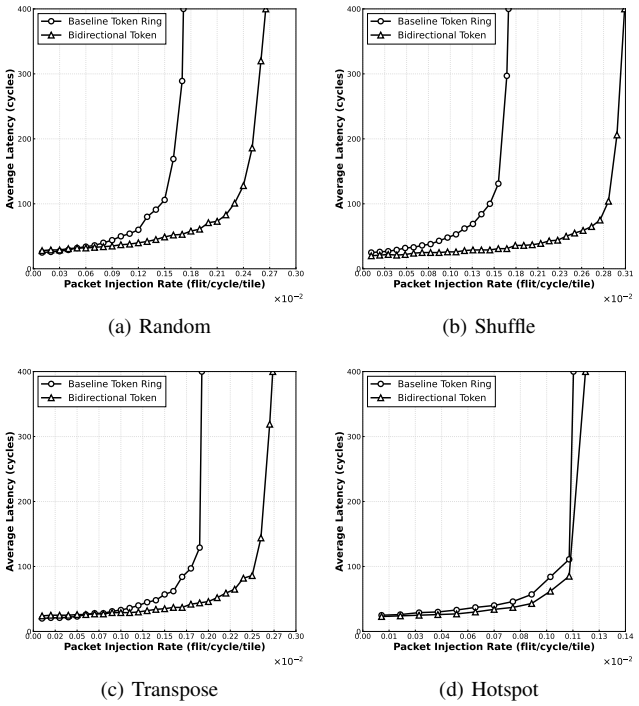


Fig. 1. Evaluated mesh-WiNoC: 64 cores share 16 radio hubs connected by bidirectional token-control links.

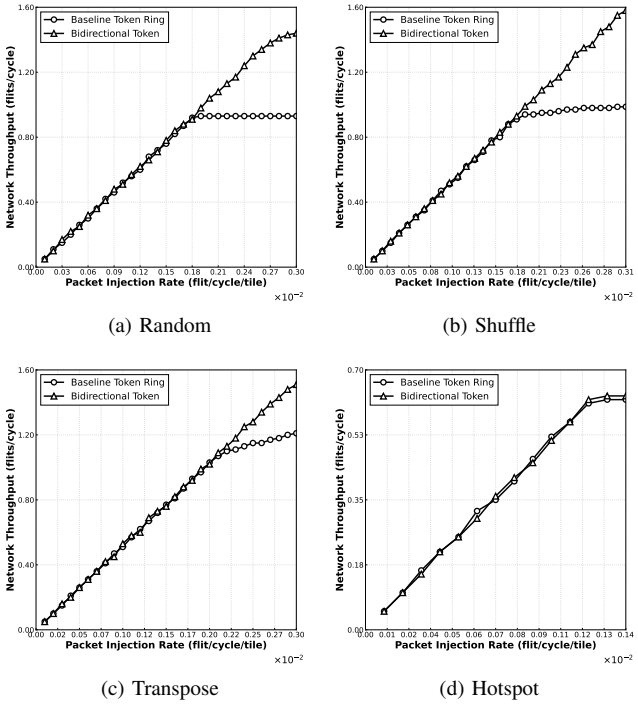


(a) Random (b) Shuffle

(c) Transpose (d) Hotspot

Fig. 2. Average packet latency versus injection rate for BMAC and the token-ring baseline under (a) random, (b) shuffle, (c) transpose, and (d) hotspot traffic.

to receive packets than other destinations. Both protocols approach saturation at similar injection rates. At the reported saturation PIR, BMAC has a latency of 84 cycles, compared with 111 cycles for the baseline, a reduction of approximately 24%. Concentrated demand can thus limit throughput even when token access changes reduce latency.



(a) Random (b) Shuffle

(c) Transpose (d) Hotspot

Fig. 3. Delivered throughput versus injection rate for BMAC and the token-ring baseline under (a) random, (b) shuffle, (c) transpose, and (d) hotspot traffic.

Throughput measures the rate at which the network delivers data at a given offered load. Figure 3 compares this rate for BMAC and the baseline [21]. As injection increases, throughput eventually approaches the capacity supported by each configuration.

The supplied results indicate throughput gains of approximately 60% for random traffic, 80% for shuffle traffic, and 36% for transpose traffic. Hotspot throughput differs little between the protocols. These observations suggest that changing token traversal is most useful when the access order contributes materially to the bottleneck. When demand is concentrated at one destination, reducing token travel alone offers limited relief.

BMAC requires links that support token transfer in either direction and a controller that decides when to return to the preceding hub. The results demonstrate a performance benefit for the evaluated mesh and workloads. They do not quantify controller area, implementation cost, or energy consumption, and they do not establish performance for larger networks.

Figure 4 summarizes the throughput comparison across the four workloads. The random, shuffle, and transpose cases favor BMAC, while the hotspot case shows only a small difference. The plotted values provide a workload-level summary of this configuration; they do not establish general scalability or quality-of-service guarantees. Those issues remain relevant to wider NoC evaluation [22], [23].

Figure 5 expresses throughput as a ratio to the baseline. The reported ratios are approximately $1.6\times$ for random traffic and $1.8\times$ for shuffle traffic, consistent with the

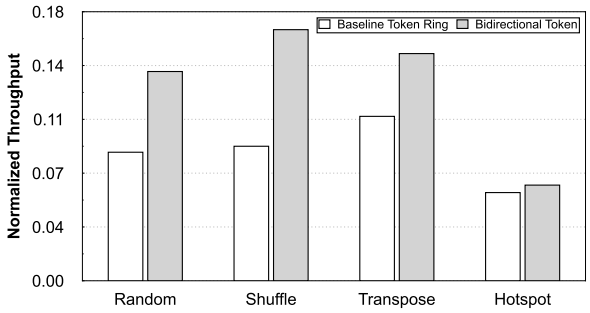


Fig. 4. Normalized network throughput comparison for the baseline and BMAC designs in the mesh-WiNoC architecture under synthetic traffic workloads.

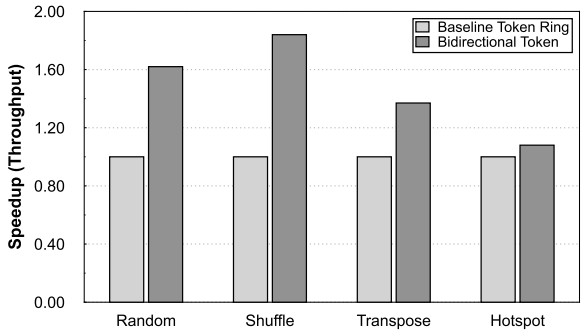


Fig. 5. Performance speedup (throughput) comparison for the baseline and BMAC designs in the mesh-WiNoC architecture under synthetic traffic workloads.

approximate percentage gains discussed above. The hotspot ratio remains close to unity, consistent with the limited throughput separation in Figure 3(d).

IV. Conclusion

Local token backtracking changes the order in which radio hubs receive wireless access. In the evaluated 64-core mesh, BMAC increases throughput under random, shuffle, and transpose traffic, with the largest reported ratio approximately $1.84\times$. Under hotspot traffic, throughput remains close to the baseline, although the reported latency falls from 111 to 84 cycles at the saturation PIR. The benefit therefore depends on where requests accumulate and how they relate to the token's position.

The analytical model illustrates the potential of shorter-direction traversal, but its ideal bounds do not constitute guarantees for the local backtracking rule. Further work should evaluate the controller's fairness behavior under sustained contention and quantify energy consumption and hardware cost. Evaluation across larger meshes, application traces, and alternative topologies would clarify the range of workloads for which the mechanism is useful.

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